

## CHALLENGE – 11 – CRYPTOGRAPHY

basic-mod2 

Medium

Cryptography

picoCTF 2022

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## Description

A new modular challenge!

Download the message [here](#).

Take each number mod 41 and find the modular inverse for the result. Then map to the following character set: 1-26 are the alphabet, 27-36 are the decimal digits, and 37 is an underscore.

Wrap your decrypted message in the picoCTF flag format (i.e.

`picoCTF{decrypted_message}`)

Hints 

1 2 3

It's recommended to use a tool to find the modular inverses

17,242 users solved



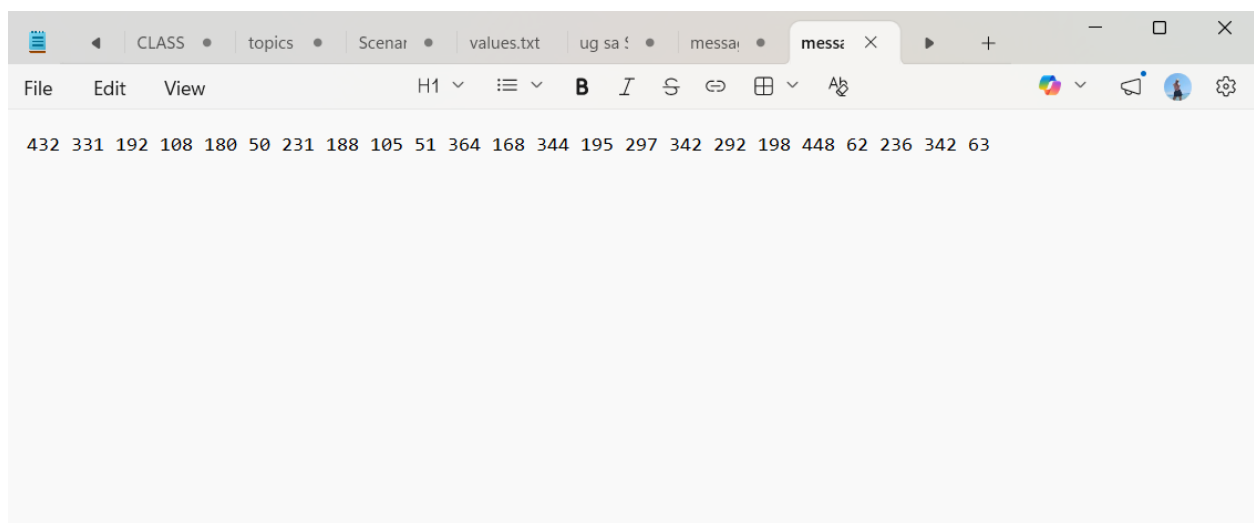
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## Description

This modular cryptography challenge requires processing a set of numbers through a specific mathematical transformation to reveal a hidden message.



## The rules for decryption are:

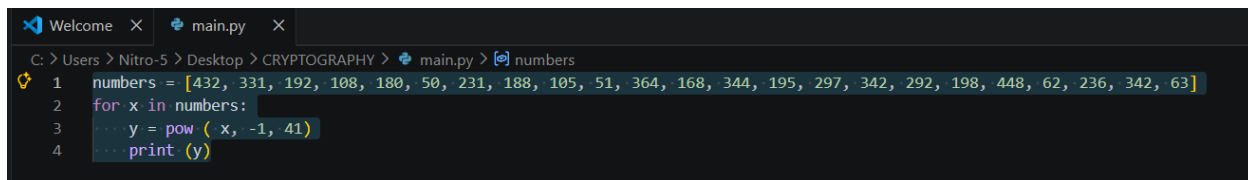
1. Take each provided number **mod 41**.
2. Find the **modular inverse** ( $x^{-1} \pmod{41}$ ) for the result.
3. Map the resulting integers to a specific character set:
  - **1-26**: The alphabet (A-Z).
  - **27-36**: Decimal digits (0-9).
  - **37**: An underscore (\_).

## Solution Approach

### 1. Data Collection

The challenge provides a list of numbers to be decrypted:

432, 331, 192, 108, 180, 50, 231, 188, 105, 51, 364, 168, 344, 195, 297, 342, 292, 198, 448, 62, 236, 342, 63



```
Welcome x main.py x
C: > Users > Nitro-5 > Desktop > CRYPTOGRAPHY > main.py > numbers
1 numbers = [432, 331, 192, 108, 180, 50, 231, 188, 105, 51, 364, 168, 344, 195, 297, 342, 292, 198, 448, 62, 236, 342, 63]
2 for x in numbers:
3     y = pow(x, -1, 41)
4     print(y)
```

### 2. Calculating Modular Inverses

Using a Python script, the modular inverse for each number was calculated using the `pow(x, -1, 41)` function. This produced the following sequence of values:

28, 14, 22, 30, 18, 32, 30, 12, 25, 37, 8, 31, 18, 4, 37, 3, 33, 35, 27, 2, 4, 3, 28

### 3. Character Mapping

The final step involved writing a decoding function to map the processed numbers back to characters:

- **Alphabet Mapping**: `chr(ord('A') + n - 1)` for values 1-26.
- **Digit Mapping**: `str(n - 27)` for values 27-36.
- **Underscore**: `_` for the value 37.

### 4. Flag Recovery

Running the completed Python script processed the sequence and output the following plaintext:

1NV3R53LY\_HARD\_C680BDC1

```
Welcome  main.py  X
C: > Users > Nitro-5 > Desktop > CRYPTOGRAPHY > main.py > ...
1  nums = [
2      28, 14, 22, 30, 18, 32, 30, 12, 25, 37,
3      8, 31, 18, 4, 37, 3, 33, 35, 27, 2,
4      4, 3, 28
5  ]
6
7  def decode(n):
8      if 1 <= n <= 26:
9          return chr(ord('A') + n - 1) # A-Z
10     elif 27 <= n <= 36:
11         return str(n - 27) # 0-9
12     elif n == 37:
13         return '_'
14     else:
15         return '?'
16
17  decoded = ''.join(decode(n) for n in nums)
18  print(decoded)

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
4
3
28
PS C:\Users\Nitro-5\Desktop\CRYPTOGRAPHY> ^C
PS C:\Users\Nitro-5\Desktop\CRYPTOGRAPHY>
PS C:\Users\Nitro-5\Desktop\CRYPTOGRAPHY> c:; cd 'c:\Users\Nitro-5\Desktop\CRYPTOGRAPHY'
1NV3R53LY_HARD_C680BDC1
PS C:\Users\Nitro-5\Desktop\CRYPTOGRAPHY> 
```

**Final Flag:** picoCTF{1NV3R53LY\_HARD\_C680BDC1}